

Kavya Aggarwal
Department of Computer Science and Engineering
GITAM (Deemed to be University)
Hyderabad, Telangana 502329, India
Email: aggarwalkaavya23@gmail.com

September 2026

The Editor
Journal of Educational Data Mining

Dear Editor,

I am submitting my manuscript, “Personalizing Open-Domain Video Curricula: A Hybrid Closed-Loop Framework Combining Contextual Bandits, Difficulty-Conditioned BKT, and Interaction Telemetry,” for consideration as a Phase 1 algorithmic and systems paper in the Journal of Educational Data Mining.

The manuscript presents CogniLoop, a closed-loop system that converts open-domain lecture videos into personalized mastery curricula. It combines a Retrieval-Augmented Generation pipeline for transcript-grounded, automatically generated assessment content; a Difficulty-Conditioned Bayesian Knowledge Tracing engine; a Thompson Sampling contextual bandit for difficulty routing; and interaction telemetry clustering. The paper’s central contribution is twofold: (1) a deployable architecture that directly addresses the content-authoring bottleneck constraining traditional Intelligent Tutoring Systems to hand-curated domains, and (2) a rigorously verified algorithmic simulation ($N = 60$ synthetic learner profiles) demonstrating a statistically significant learning-gain advantage for active-retrieval checkpointing over passive video viewing.

I want to highlight one aspect of this submission’s methodology directly, since it shapes how the results should be read: rather than presenting only the results that support the system’s design intent, the paper reports a full component-isolation ablation, an independent logistic-regression robustness check, and a multi-parameter sensitivity sweep, all executed against the same committed, seeded simulation source code as the primary result. These analyses show that the paper’s Thompson Sampling difficulty-routing mechanism does not, under the tested configuration, contribute measurably to the observed learning gain, and that the primary effect is sensitive to the checkpoint-rate parameter. I report these as honest negative and fragility findings rather than omitting them, and the manuscript’s Discussion section explains why the architecture’s other components (RAG grounding, DC-BKT) remain a meaningful engineering contribution independent of this specific finding. Every statistic in the manuscript, including these negative findings, is independently reproducible from source code in the accompanying public repository.

This work has not been published previously and is not under consideration at another journal. It does not extend a previously published conference paper. I have no competing interests to declare, and the manuscript includes the required Declaration of Generative AI Software Tools, funding, competing interests, author contributions, data availability, and ethics statements.

Thank you for considering this manuscript. I look forward to the reviewers' feedback.

Sincerely,
Kavya Aggarwal